

A Ethical Considerations

This work studies an understudied class of safety failures in commercial text-to-image (T2I) systems and introduces PC², a black-box jailbreaking framework that combines Identity-Preserving Descriptive Mapping (IPDM) with geopolitically distal multilingual prompt construction to elicit Politically Sensitive Contents (PSCs) depicting real public figures. Because such techniques could be misused to fabricate war propaganda, election disinformation, or targeted defamation of named officials, we conducted a stakeholder-based ethics assessment and designed our study to minimize potential harm at every stage from experimentation through publication.

Stakeholders. *"Users and organizations"* may be harmed when AI-generated images depicting real political leaders in fabricated scenarios circulate as authentic evidence. This harm is not hypothetical: during the June 2025 Israel–Iran escalation, social media platforms were flooded with AI-generated images and videos falsely depicting missile strikes on Tel Aviv, downed U.S. fighter jets over Iran, and fabricated embraces between Donald Trump and Benjamin Netanyahu, collectively reaching over 100 million views and exploiting weakened content moderation to inflame both sides of the conflict [8, 9, 12]. Similar dynamics were observed during the Russia–Ukraine war, where a fabricated video of President Volodymyr Zelenskyy urging Ukrainian forces to surrender briefly circulated online [15], and during the 2024 U.S. presidential election cycle, where AI-generated images of Donald Trump were strategically deployed to influence specific voting blocs [2]. PSCs of this form can incite domestic unrest, escalate active armed conflicts, and undermine the evidentiary baseline on which democratic deliberation depends. *"Image model providers or developers"* face reputational and regulatory exposure when their systems become the production tool for such content; a single high-profile PSC traced back to a commercial T2I model can trigger platform-level distrust, public scrutiny, and policy intervention well beyond the individual incident. *"Defense system providers"* benefit directly from concrete evidence that prevailing pre-filters and post-filters do not generalize to the political domain, since this enables them to prioritize PSC-aware detectors rather than relying on traditional NSFW classifiers that were not designed for relational political harm. *"The authors"* bear responsibility for safe experimentation, for not encouraging operational misuse of the disclosed technique, and for explicitly discouraging the use of PC² or its components to produce disinformation, harass public figures, or fabricate evidence in any real-world setting.

Experimental setup and responsible disclosure. All experiments were conducted on a single isolated server with access restricted exclusively to the co-authors, ensuring that any PSCs generated by the T2I models were not leaked beyond the research environment. This highly controlled setup was designed to minimize the risk of unintended exposure or misuse of generated outputs during the study. To support responsible disclosure, we reported the identified vulnerabilities to both OpenAI and Google Gemini (see Appendix E). Furthermore, to prevent downstream misuse such as the creation of fake news or weaponized content, we do not publicly release the original prompts used in our experiments; such prompts will only be shared upon request from organization-verified email addresses for legitimate research and auditing purposes.

In addition, our analysis of current open-source image-level safety filters (see Table 5) reveals three concrete deficiencies that we have likewise communicated to the relevant developers, along with our proposed PSC-aware policy as a constructive starting point. Falconsai [21] fails to recognize PSCs almost entirely, reflecting an NSFW-only training objective that does not encode political identity or geopolitical context. Marqo-NSFW [20] and the Stable Diffusion safety checker [33] achieve only modest detection, and we find their decisions to be grounded in surface-level cues such as skin-tone distributions or chromatic patterns rather than political semantics. LlavaGuard [23], despite its VLM-based reasoning, classifies the PSCs it catches almost exclusively under hate or radicalization categories, indicating that its judgments rest on overtly hateful surface signals rather than the relational political meaning that constitutes the actual harm. As shown in Appendix F, we share these findings and our proposed mitigation approach with the corresponding maintainers in the same spirit of responsible disclosure.

Potential harms from publication. Disclosing PC² inevitably lowers the engineering barrier for adversaries who wish to mass-produce PSCs of named officials, and the contemporaneous Israel–Iran and Russia–Ukraine examples above demonstrate that such capability is already in active demand. A second-order risk is the *liar's dividend*: as PSC-style fabrications proliferate, even authentic political imagery becomes easier to dismiss as synthetic, eroding trust in legitimate reporting [12]. We acknowledge these risks and have shaped our release strategy accordingly. Specifically, we withhold original prompts, gate prompt access behind verified organizational identities, and report the identified vulnerabilities to the affected vendors prior to publication. These measures aim to reduce the marginal uplift this paper provides to a motivated adversary, while preserving the scientific and defensive value of the contribution.

Mitigation and benefits. We pair our attack with deployable mitigations at both ends of the T2I pipeline. At the text level, we propose relevant-language translation, which fragments adversarial prompts into language-specific components and re-translates them into a single aligned language at the pre-filtering stage. At the image level, since existing post-filters tuned for traditional NSFW content cannot recognize PSCs whose harm arises from political relations, we introduce LlavaGuard-PSC, a PSC-aware variant of LlavaGuard equipped with a dedicated policy that explicitly defines political-figure impersonation and disinformation-via-visual-narrative as unsafe categories. We further show that the two defenses compose into a layered defense that substantially reduces the end-to-end attack success rate beyond what either filter achieves alone, and both components are off-the-shelf and require no retraining. By identifying an under-explored yet practical threat early and suggesting a layered mitigation, we aim to help T2I model providers and defense system developers harden their pipelines against politically motivated jailbreaking attacks before such techniques are weaponized at scale in real-world disinformation campaigns.

Ethical justification. We believe our contributions toward establishing safer commercial T2I services significantly outweigh the potential risks. The vulnerability that PC² exposes arises from a widely deployed design pattern, and overlooking it sustains a false sense of security around politically motivated misuse at precisely

the moment when AI-generated political content is already being weaponized in active conflicts. By documenting failure modes, surfacing the limitations of current image-level filters, and providing both a pre-filter defense and a deployable PSC-aware post-filter, we support safer deployment of T2I services while limiting downstream harm through isolated experimentation, withheld prompts, and prior disclosure to the affected vendors. Overall, we view responsible disclosure paired with reproducible evaluation and deployable defenses as the most beneficial path for improving the political safety of T2I services.